Derek Umoren

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Technical Skills

Languages: TypeScript, JavaScript, Python, Go, C/C++, SQL
Frameworks: React, Next.js, Vue 3, Nuxt 3, FastAPI, Node.js, .NET 8
Data & Infrastructure: PostgreSQL, Redis, AWS, Docker, Kubernetes, GitHub Actions
Testing & Observability: pytest, Playwright, axe-core, Prometheus, Grafana

Education

Memorial University of Newfoundland Expected May 2027
*B.Eng. in Computer Engineering (**GPA: 3.76/4.00**); Dean's List 2022–24* *St. John's, NL*

Experience

Angler Solutions Inc. St. John's, NL
Software Engineering Intern *Jan 2026 – Apr 2026*

- Engineered a full-stack LOHC hydrogen transportation simulator with **Python, NumPy, Pandas, FastAPI, Nuxt 3, Vue 3, and TypeScript**, processing **8,760-hour** renewable-energy profiles across **4 electrolyzer configurations**.
- Converted Excel models for **3 clean-energy systems (LOHC, ACAES, and DCAES)** into production Python and **AWS Lambda/SAM**.
- Automated **8+ financial KPIs**, including LCOH/LCOE/LCOS, NPV, IRR, and payback, across the clean-energy simulation workflows.
- Authored **181 pytest unit and regression tests** covering **7 LOHC model components**, hourly dispatch, hydrogen production, GHG emissions, and financial calculations.
- Added structured diagnostics across **FastAPI, AWS Lambda, and S3-backed workflows** to improve simulation troubleshooting.
- Built **18 interactive ECharts visualizations** and CSV/XLSX exports in **Nuxt/Vue** for simulation analysis.
- Standardized dark-mode image exports across **40 visualization components**, improving consistency in generated reports.

Software Engineering Intern *May 2025 – Aug 2025*

- Built full-stack simulation features with **Nuxt.js, FastAPI, Docker, and PostgreSQL**.
- Accelerated deployments through **AWS Elastic Beanstalk** and GitHub CI workflows.
- Optimized the wind-speed data pipeline, cutting simulation runtime by **65%** and preparing the architecture for scalable public API access.
- Trained and deployed ML models for wind-speed imputation with sub-second prediction latency.

Bluedrop ISM St. John's, NL
Full Stack Software Developer Intern *Sep 2024 – Dec 2024*

- Implemented **pgTyped** to generate TypeScript types directly from SQL, strengthening compile-time safety, improving code reliability, and reducing database integration errors.
- Moved database queries and logic from the shared data-access layer into feature-aligned functions, simplifying architecture and improving maintainability and efficiency by **20%**.
- Integrated **GraphQL** queries and mutations into the front end to replace RESTful APIs, improving data-fetching efficiency and enabling flexible client-server interaction.
- Used **Docker** and **Kubernetes** to test and manage containerized environments for smoother development and deployment.

Projects

A11yGuard | *TypeScript*, *Playwright*, *axe-core*, *BullMQ*, *PostgreSQL*

- Built a GitHub App that scans pull-request previews, compares findings against the main-branch baseline, and reports new WCAG regressions through GitHub Checks.
- Created a deterministic and LLM-guided Playwright explorer with reproducible state graphs; BullMQ/Redis workers replay and rescan candidate fixes against PostgreSQL-backed history.

GreenQueue | *Go*, *Kubernetes*, *.NET 8*, *Prometheus*

- Built a deadline-aware Kubernetes controller in Go that shifts batch Jobs to lower-carbon periods while honoring runtime, safety-margin, and deadline constraints.
- Integrated a .NET 8/OpenAPI Carbon Aware SDK adapter and deterministic replay for 4 scheduling policies; validated with 300+ Go, 38 .NET, and 21 Kind E2E tests plus Prometheus/Grafana.